# Project 1 – Evaluation of Indic Translation Models

*Round-2 Hands-On Project · Tamizh Internship*

## 1. Executive Summary

This project evaluates English → Tamil translation across three complementary dimensions: translation quality (sacreBLEU), comparative token-based analysis across five tokenizers, and Indic token behaviour focused on how Tamil's agglutinative morphology interacts with each model's tokenizer. IndicTrans2 achieves a sacreBLEU score of 22.42 on 100 FLORES-200 sentences (the "good" band for English → Tamil). NLLB-600M produces the most token-efficient valid Tamil output. mT5-base, although the most compact at the token level, does not produce valid Tamil because it is not fine-tuned for machine translation.

## 2. Objective

Quantitatively compare modern English → Tamil translation models on a shared dataset, and analyse how their tokenizers behave on Indic (Tamil) text, in order to inform model selection for downstream Tamil NLP work.

## 3. Dataset

FLORES-200 devtest split, the human-translated multilingual evaluation set released by Meta AI. The repository ships a merged CSV at `data/raw/translation_dataset.csv` with 1012 sentences and the columns `id`, `english`, `tamil_reference`, `domain`, `topic`. Part A uses the first 100 rows (predominantly wikinews domain due to row ordering); Parts B and C use the first 10 rows so that the analysis is dominated by tokenizer behaviour rather than corpus variance.

## 4. Models Used

| Short name | Hugging Face ID | Used in |
|---|---|---|
| IndicTrans2 | ai4bharat/indictrans2-en-indic-1B | Parts A, B, C |
| NLLB-600M | facebook/nllb-200-distilled-600M | Parts B, C |
| mT5-base | google/mt5-base | Parts B, C |
| OpusMT-mul | Helsinki-NLP/opus-mt-en-mul | Parts B, C |
| MADLAD-3B | google/madlad400-3b-mt | Parts B, C |

Model substitution notes (deviations from the submission guide):

- `Helsinki-NLP/opus-mt-en-ta` is not a public model on Hugging Face. The repository substitutes `Helsinki-NLP/opus-mt-en-mul` (English → multilingual), which supports Tamil as a target and shares the same MarianMT tokenizer architecture, so it is valid for tokenizer-level analysis.
- `google/madlad400-1b-mt` is not publicly available; the smallest public MADLAD variant `google/madlad400-3b-mt` is used instead.

# 5. Methodology

## 5.1 Part A – batch translation & sacreBLEU

IndicTrans2 (1B parameters) is loaded once from Hugging Face and used together with the AI4Bharat IndicProcessor preprocessor, which adds the source tag `eng_Latn` and target tag `tam_Taml` and performs the internal romanization / post-processing the model expects. The first 100 English sentences are translated in batches; predictions and the human Tamil reference are scored with the sacreBLEU library to produce a corpus BLEU score, brevity penalty, and length ratio. Outputs are saved to `translation_outputs.csv` and `sacrebleu_results.csv` under `part_a_batch_translation/`.

## 5.2 Part B – token-based comparative analysis

Each of the 5 models translates the first 10 English sentences; the corresponding tokenizer is then used to count tokens on both the source and the produced Tamil output. For every (sentence, model) pair the notebook records `source_token_count`, `target_token_count`, `expansion_ratio = target / source`, `avg_word_length`, `subword_fragmentation` and `unknown_token_rate`. The EDA includes a histogram of token counts, a boxplot of expansion ratios, a scatter of source vs. target tokens and a bar chart of average expansion per model.

## 5.3 Part C – Indic token behaviour

Using the same 5 tokenizers, the notebook isolates Tamil-script tokens and computes: average characters per Tamil token, Unicode fragmentation (share of tokens that are a single Tamil code point), rare-token rate, vocabulary coverage on Tamil text, and an illustrative embedding-memory footprint, assuming a fixed embedding hidden size of 768 (memory ≈ vocab_size × 768 × 4 bytes). Plots cover all five metrics.

## 6. Part A – Translation Quality Results

| Metric | Value |
|---|---|
| Model | ai4bharat/indictrans2-en-indic-1B |
| Sentences scored | 100 |
| SacreBLEU | 22.4231 |
| Brevity Penalty | 0.9638 |
| Length Ratio | 0.9645 |

A sacreBLEU score of 22.42 sits inside the typical "good" band for English → Tamil (reasonable: 5–15, good: 15–25, excellent: 25+). Brevity penalty (0.9638) and length ratio (0.9645) close to 1.0 indicate IndicTrans2 produces Tamil outputs of roughly the same length as the human reference, with no systematic under-generation.

## 7. Part B – Token-Based Comparative Analysis

| Model | Avg expansion (tgt / src) | Valid Tamil output |
|---|---|---|
| IndicTrans2 | 5.256 | Yes |
| NLLB-600M | 1.366 | Yes – best balance |
| MADLAD-3B | 1.641 | Yes |
| OpusMT-mul | 2.421 | Yes |
| mT5-base | 0.200 | No – not fine-tuned for MT |

Among models that produce valid Tamil, NLLB-600M has the lowest expansion ratio (1.366). IndicTrans2's apparent expansion of 5.256 is inflated because the IndicProcessor preprocessing layer inserts romanized Indic forms and language tags that its tokenizer counts as additional tokens (see §9.3). mT5-base produces almost no Tamil at all because the base mT5 checkpoint is pre-trained but not fine-tuned for translation.

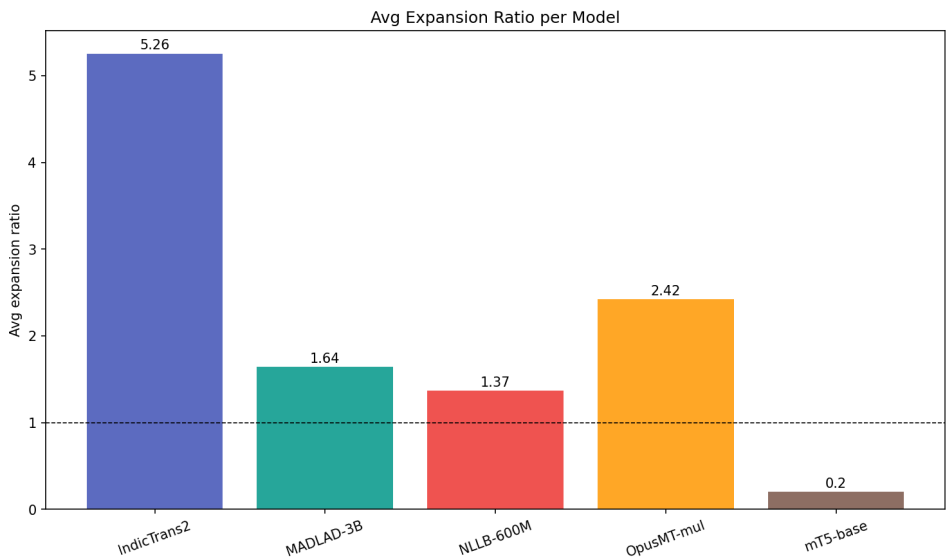


*Figure 1. Average target / source token expansion per model.*

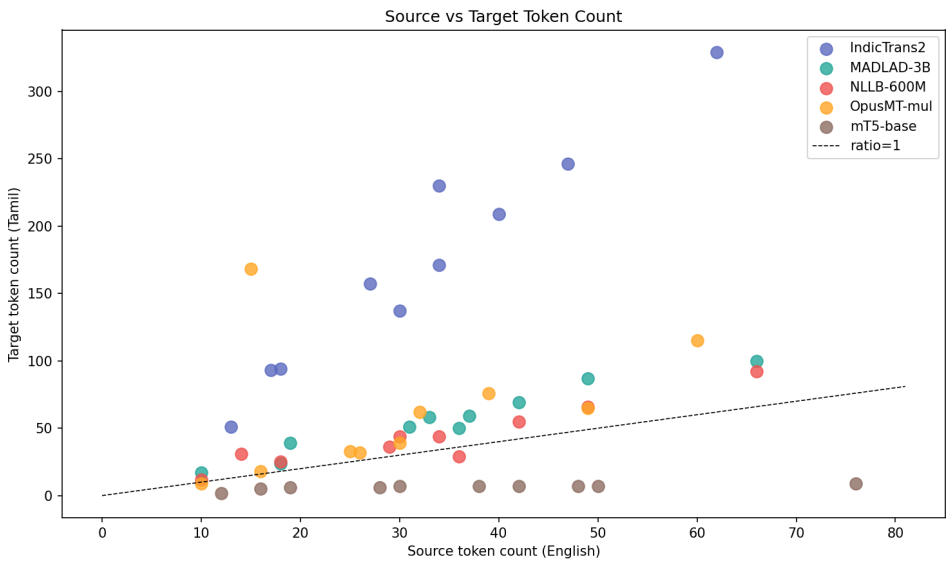


*Figure 2. Source vs. target token counts (per sentence, per model).*

## 8. Part C – Indic Token Behaviour

| Model | Avg chars / token | Unicode frag. | Memory* (MB) |
|---|---|---|---|
| IndicTrans2 | 1.004 | 0.861 | 102.268 |
| NLLB-600M | 3.035 | 0.241 | 9.093 |
| mT5-base | 3.097 | 0.148 | 8.825 |
| OpusMT-mul | 0.929 | 0.850 | 96.134 |
| MADLAD-3B | 2.678 | 0.237 | 11.612 |

**Memory is an illustrative estimate from* `vocab_size × hidden_size(768) × 4 bytes`*. Useful for relative comparison only; not production model memory.*

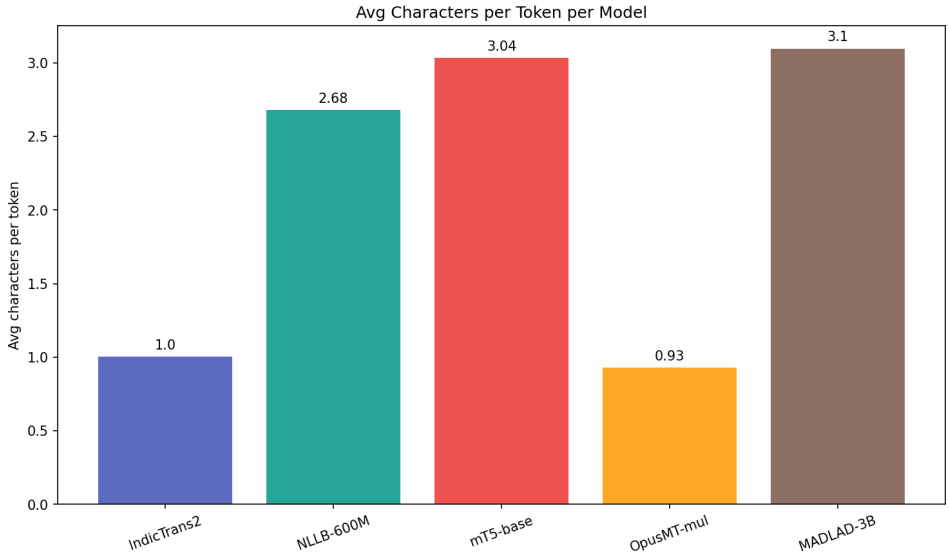


*Figure 3. Average characters per Tamil token.*

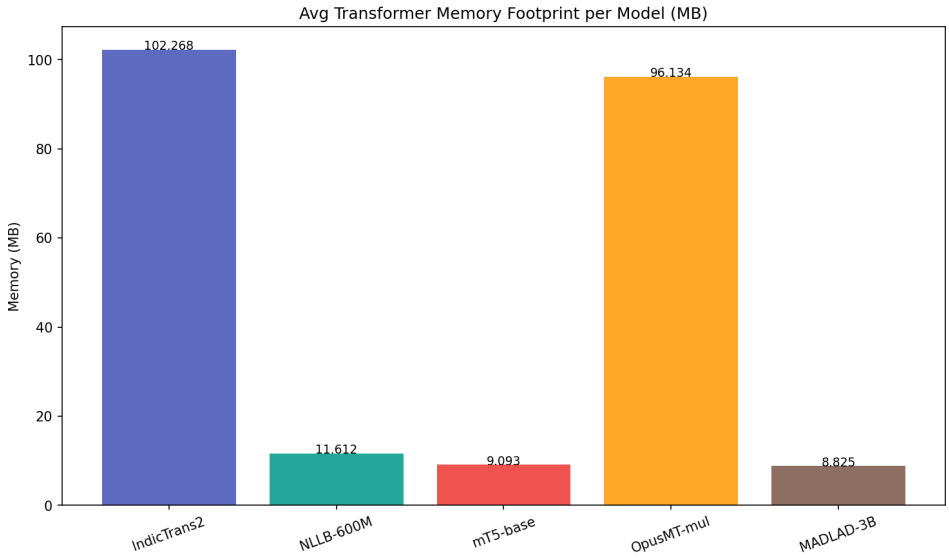


*Figure 4. Illustrative embedding-memory footprint.*

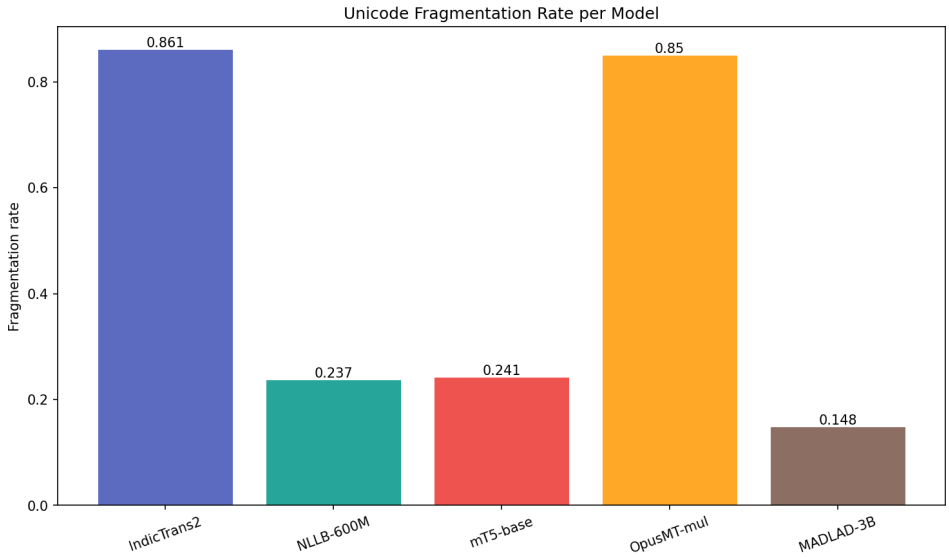


*Figure 5. Unicode fragmentation rate on Tamil text.*

# 9. Discussion

## 9.1 Why Tamil expands more than English

Tamil is agglutinative: a single English word frequently maps to a Tamil word that bundles tense, case, number and politeness suffixes into one surface form. Tokenizers trained predominantly on European text do not learn long Tamil sub-word units, so Tamil words break into many small pieces. This is the structural reason for the consistently higher target token counts seen across all five models in Part B.

## 9.2 SentencePiece vs. BPE

NLLB-600M, mT5-base and MADLAD use SentencePiece, which operates directly on raw Unicode and can keep multi-character Tamil graphemes together when the training corpus exposes them — hence the high chars / token (around 3) and the low fragmentation (around 0.15–0.24). OpusMT-mul uses a Marian BPE built largely on Latin-script bilingual corpora; with very little Tamil exposure, it falls back to character-level splits on Tamil, producing 0.929 chars / token and 85% single-codepoint tokens.

## 9.3 IndicTrans2 – high token count is a design choice

IndicTrans2 uses an IndicProcessor pre- and post-processing layer that romanizes Indic input before tokenization and transliterates back to Tamil after decoding. Externally its tokenizer therefore appears to produce many small tokens for Tamil text (chars / token ≈ 1.0, fragmentation ≈ 0.86). This is not a quality weakness: IndicTrans2 still attains the highest sacreBLEU score (22.42) in Part A. The lesson is that token metrics on the raw tokenizer can be misleading when a model wraps its tokenizer in a preprocessor.

## 9.4 Vocabulary coverage and memory cost

Models with a Tamil-aware vocabulary (NLLB, mT5, MADLAD) need only 9–12 MB of embedding memory to cover Tamil at this fixed hidden size, whereas the models that fragment Tamil (IndicTrans2 by design, OpusMT-mul by accident) need roughly 10× more — about 100 MB purely from the embedding table. At Transformer scale this token explosion also translates into higher attention compute per sentence.

## 9.5 Tokenizer efficiency is not translation quality

mT5-base ranks best on most tokenizer metrics yet fails to translate, while IndicTrans2 ranks worst on tokenizer metrics yet wins on sacreBLEU. The two evaluation axes must therefore be reported independently when selecting a model for production Indic NLP.

# 10. Caveats and Limitations

- Parts B and C use only 10 sentences, so absolute averages are small-sample and may shift on a wider corpus.
- The Memory column is illustrative only; real model memory depends on the actual hidden size, precision and architecture of each model.
- mT5-base is used as released with no MT fine-tuning; the same tokenizer with an mT5 MT fine-tune would likely behave differently.
- Two model substitutions were necessary — see §4.
- All experiments run on CPU; numbers reflect tokenizer behaviour, not GPU-accelerated inference speed.

## 11. Conclusion

For English → Tamil translation, IndicTrans2 is the strongest production baseline on this dataset (sacreBLEU 22.42). When token efficiency is the priority and quality is still required, NLLB-600M is the best balance: valid Tamil output, the lowest expansion ratio among valid-output models, and the smallest embedding-memory footprint. Tokenizer-only metrics like expansion ratio or fragmentation are necessary but not sufficient indicators of translation quality, and tokenizer numbers on models with preprocessors (e.g. IndicTrans2) must be interpreted with care.

## 12. References


- NLLB Team. No Language Left Behind: Scaling Human-Centered Machine Translation. Meta AI, 2022. (FLORES-200 dataset.)
- AI4Bharat. IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages.
- Post, M. A Call for Clarity in Reporting BLEU Scores. (sacreBLEU.)
- Kudo, T. & Richardson, J. SentencePiece: A simple and language-independent subword tokenizer.